

On the Impossibility of Autonomous Self-Improvement to General Intelligence in Stochastic Language Models: A Proof Program via Information Theory, Gödelian Limits, and Distributional Collapse

Revised Edition with Counterargument Analysis

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Abstract

We develop a proof program for the claim that large language models (LLMs) based on stochastic next-token prediction cannot, through autonomous self-improvement without external human influence, converge to artificial general intelligence (AGI). We model the LLM as a conditional probability estimator trained via maximum likelihood on a fixed distributional support and identify independent barriers requiring formal verification: (1) the information-theoretic ceiling imposed by training data entropy; (2) distributional collapse under recursive self-training; (3) Gödelian and computability bounds on self-verification; (4) the stochastic gradient descent fixed-point trap; (5) the gap between statistical pattern completion and general reasoning; (6) conditional computational-complexity barriers; (7) context and computation resource constraints; and (8) error compounding under recursive self-modification. We address counterarguments regarding inference-time search, reinforcement learning with self-play, tool-augmented systems, and multi-agent architectures, arguing that each either falls within the proof's scope or introduces external information that should be treated as human influence under the definitions below. We further provide a control theory perspective and empirical corroboration from model-collapse, self-verification, and code-maintenance benchmarks. The current Lean formalization is a work in progress: several bridge assumptions and theorem obligations remain open, so the conclusion should be read as a conditional proof program rather than a completed machine-checked theorem.

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1 Scope and Definitions

We begin by precisely delineating what this proof does and does not claim.

Remark 1.1 (Formalization Status). *This manuscript is accompanied by a Lean 4 formalization skeleton. The Lean project currently typechecks, but it is not yet a completed proof: several central results are represented by proof obligations, bridge axioms, or placeholders. The claims below should therefore be read in three categories:*

1. Formalized components, *such as selected computability facts imported from Mathlib;*
2. Conditional theorem targets, *whose conclusions depend on assumptions that must still be discharged;*
3. Empirical or philosophical support, *which may motivate the theorem targets but does not substitute for formal proof.*

The verification plan for closing these gaps is maintained separately in docs/verification_plan.md.

Remark 1.2 (Scope). *This proof program addresses the following specific question: Can a system based on autoregressive stochastic language modeling, trained via maximum likelihood estimation (or equivalent objectives), autonomously self-improve to AGI without any form of human influence at any stage of the self-improvement process?*

*We do **not** claim:*

- *That AGI is impossible by any means whatsoever.*
- *That hybrid systems combining LLMs with other components cannot advance.*
- *That human-guided improvement of LLMs is bounded.*

*We **do** target the following conditional claim: within the stochastic autoregressive framework operating autonomously, the mathematical structure of the system prevents convergence to AGI once the stated barriers are fully formalized. Any system that escapes these bounds should be analyzed by asking whether it incorporates elements that constitute human influence or departs from the stochastic language modeling paradigm.*

1.1 The Large Language Model as a Stochastic System

Definition 1.3 (Autoregressive Language Model). *An autoregressive language model is a parameterized conditional probability distribution over a vocabulary $\mathcal{V} = \{v_1, v_2, \dots, v_V\}$:*

$$p_{\boldsymbol{\theta}}(x_t \mid x_{<t}) = \text{softmax}(\mathbf{W}_{\text{out}} \cdot h_L(\mathbf{x}_{<t}; \boldsymbol{\theta}))_{x_t} \quad (1)$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is the parameter vector (with p on the order of 10^9 to 10^{12}), and h_L is the output of an L -layer transformer network. The full sequence probability factorizes as:

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{t=1}^T p_{\boldsymbol{\theta}}(x_t \mid x_{<t}) \quad (2)$$

Definition 1.4 (Transformer Computation). *Each transformer layer $\ell \in \{1, \dots, L\}$ computes:*

$$\mathbf{z}^{(\ell)} = \text{LayerNorm}\left(\mathbf{h}^{(\ell-1)} + \text{MultiHeadAttn}(\mathbf{h}^{(\ell-1)})\right) \quad (3)$$

$$\mathbf{h}^{(\ell)} = \text{LayerNorm}\left(\mathbf{z}^{(\ell)} + \text{FFN}(\mathbf{z}^{(\ell)})\right) \quad (4)$$

where multi-head attention with H heads is:

$$\text{MultiHeadAttn}(\mathbf{h}) = \mathbf{W}_O \left[\bigoplus_{i=1}^H \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k}} \right) \mathbf{V}_i \right] \quad (5)$$

with $\mathbf{Q}_i = \mathbf{W}_Q^{(i)} \mathbf{h}$, $\mathbf{K}_i = \mathbf{W}_K^{(i)} \mathbf{h}$, $\mathbf{V}_i = \mathbf{W}_V^{(i)} \mathbf{h}$, and:

$$\text{FFN}(\mathbf{z}) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2 \quad (6)$$

for activation function σ (typically GELU or SiLU).

Definition 1.5 (Training Objective). *The model is trained to minimize the cross-entropy loss:*

$$\mathcal{L}(\boldsymbol{\theta}) = -\mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\text{train}}} \left[\frac{1}{T} \sum_{t=1}^T \log p_{\boldsymbol{\theta}}(x_t \mid x_{<t}) \right] = H(p_{\text{data}}) + \text{KL}(p_{\text{data}} \parallel p_{\boldsymbol{\theta}}) \quad (7)$$

1.2 Defining Autonomous Self-Improvement

Definition 1.6 (Human Influence). *We define human influence as any of the following introduced at step $k \geq 1$:*

1. *New human-curated or human-generated training data,*
2. *Human-designed reward functions, preference labels, or evaluation criteria,*
3. *Human-designed filtering, selection, or ranking mechanisms applied to model outputs,*
4. *Human-designed tools, environments, or external verifiers (including game rules, physics engines, code interpreters, proof checkers),*
5. *Human-designed architectural modifications or hyperparameter schedules.*

Note that items (2)–(4) include systems where humans designed the evaluation mechanism before the self-improvement loop begins. A reward function crafted by humans is human influence even if no human is present during training. This is the critical distinction: the intelligence embedded in the design of the feedback signal originates from human cognition.

Definition 1.7 (Autonomous Self-Improvement). *An autonomous self-improvement process is a sequence of models $\{\boldsymbol{\theta}_k\}_{k=0}^\infty$ where:*

$$\boldsymbol{\theta}_{k+1} = \mathcal{A}(\boldsymbol{\theta}_k, \mathcal{D}_k^{\text{self}}) \quad (8)$$

where $\mathcal{A}$ is a training algorithm and $\mathcal{D}_k^{\text{self}}$ is data generated by the model:

$$\mathcal{D}_k^{\text{self}} = \left\{ \mathbf{x}^{(i)} \sim p_{\boldsymbol{\theta}_k}(\cdot) \right\}_{i=1}^{N_k} \quad (9)$$

with **no human influence** (Definition 1.6) at any step $k \geq 1$.

Definition 1.8 (Artificial General Intelligence — Operational). *A system achieves AGI if there exists a model $\boldsymbol{\theta}^*$ such that for any computable function $f : \Sigma^* \rightarrow \Sigma^*$ and any $\epsilon > 0$:*

$$\Pr_{\mathbf{x} \sim \mathcal{D}_f} [p_{\boldsymbol{\theta}^*}(\cdot \mid \mathbf{x}) \text{ assigns highest probability to } f(\mathbf{x})] \geq 1 - \epsilon \quad (10)$$

Assumption 1 (AGI Target Nontriviality). *The AGI criterion is interpreted over a specified family of task distributions $\mathcal{D}_f$, input encodings, and resource bounds. This assumption is necessary to avoid making the target impossible for every finite physical system by definition. A complete formalization must state these restrictions explicitly and prove that the target remains nontrivial.*

2 Information-Theoretic Ceiling

Theorem 2.1 (Conditional Information Ceiling). *Let p_{true} denote the ground-truth distribution over all valid reasoning chains. Suppose the variables representing ground truth, training data, the initial model, self-generated data, and later models form the Markov chain used below. For the autonomous self-improvement sequence $\{\theta_k\}$:*

$$I(p_{\theta_k}; p_{\text{true}}) \leq I(p_{\theta_0}; p_{\text{true}}) \leq I(\mathcal{D}_{\text{train}}; p_{\text{true}}) \quad (11)$$

Proof. The self-generated dataset $\mathcal{D}_k^{\text{self}}$ is produced by sampling from p_{θ_k} . Under the Markov-chain assumption:

$$p_{\text{true}} \rightarrow \mathcal{D}_{\text{train}} \rightarrow \theta_0 \rightarrow p_{\theta_0} \rightarrow \mathcal{D}_0^{\text{self}} \rightarrow \theta_1 \rightarrow \dots \rightarrow \theta_k \quad (12)$$

each transformation satisfies:

$$I(\theta_{k+1}; p_{\text{true}}) \leq I(\theta_k; p_{\text{true}}) \quad (13)$$

with strict inequality whenever $p_{\theta_k} \neq p_{\text{true}}$. $\square$

Remark 2.2 (Information vs. Computation). *The information ceiling is a statement about mutual information under the stated Markov model. It does not by itself rule out better extraction or use of information already present in the model. A complete proof must therefore distinguish information creation from computational access to latent information.*

Objection 1 (Inference-Time Search and Latent Information). *Chain-of-thought reasoning, beam search, and tree search can extract capabilities “latent” in the weights without adding new data. Doesn’t this violate the information ceiling?*

Response 1. *No. Inference-time search reorganizes and better utilizes information already present in θ_k . This is analogous to lossless compression: you can rearrange bits to make them more useful, but you cannot create bits that were not there. Formally, let $\mathcal{S}$ be any deterministic search procedure applied at inference time. Then:*

$$I(\mathcal{S}(p_{\theta_k}); p_{\text{true}}) \leq I(p_{\theta_k}; p_{\text{true}}) \leq I(\mathcal{D}_{\text{train}}; p_{\text{true}}) \quad (14)$$

by the data processing inequality (since $\mathcal{S}$ is a function of p_{θ_k}). Search can improve utilization of existing information up to a ceiling, but that ceiling is fixed by the training data. Better utilization is not the same as new knowledge. The model may solve problems it previously failed on by searching more effectively, but it cannot solve problems whose solutions require information absent from the training distribution. This is the distinction between extraction (bounded) and creation (what AGI requires for novel domains).

Empirically, Gudibande et al. [14] demonstrate that fine-tuning open-source LLMs on ChatGPT-generated data transfers surface-level stylistic patterns but fails to close capability gaps on factual and reasoning benchmarks, consistent with the data processing inequality prediction that self-generated data cannot exceed the teacher’s information content. Seddik et al. [15] provide a complementary statistical proof that model collapse is unavoidable under purely synthetic training, deriving a critical mixing threshold above which degradation is inevitable regardless of hyperparameters.

3 Distributional Collapse Under Self-Training

Theorem 3.1 (Conditional Entropy Contraction Under Self-Distillation). *Let p_{θ_k} be the model at step k , and let $p_{\theta_{k+1}}$ be obtained by training on N samples from p_{θ_k} with no external filtering,*

selection, smoothing, entropy regularization, or fresh external data. Under this finite-sample empirical self-distillation regime:

$$H(p_{\theta_{k+1}}) \leq H(p_{\theta_k}) \quad (15)$$

with strict inequality requiring the stated finite-sample and training assumptions. The empirical support contracts monotonically:

$$\text{supp}(p_{\theta_{k+1}}) \subseteq \text{supp}(p_{\theta_k}) \quad (16)$$

Proof. Step 1: Sampling prunes the tail. The empirical distribution $\hat{p}_k$ over N samples satisfies: for any sequence with $p_{\theta_k}(\mathbf{x}) < 1/N$:

$$\Pr[\mathbf{x} \notin \mathcal{D}_k^{\text{self}}] = (1 - p_{\theta_k}(\mathbf{x}))^N \geq \left(1 - \frac{1}{N}\right)^N \rightarrow e^{-1} \quad (17)$$

Step 2: Training concentrates. Training θ_{k+1} to minimize $\text{KL}(\hat{p}_k \| p_{\theta})$ drives $p_{\theta_{k+1}}$ toward $\hat{p}_k$, which has smaller support and lower entropy than p_{θ_k} .

Step 3: Induction. $H(p_{\theta_0}) \geq H(p_{\theta_1}) \geq H(p_{\theta_2}) \geq \dots \geq 0$. The model becomes progressively less diverse. $\square$

Corollary 3.2 (Loss of Generality). *AGI requires $\text{supp}(p_{\theta^*}) \supseteq \bigcup_{f \in \mathcal{F}} \text{supp}(\mathcal{D}_f)$. Since self-training contracts the support, the model moves away from AGI with each iteration.*

Objection 2 (Filtering and Reinforcement Learning). *AlphaZero improved through self-play. If a model generates 100 solutions and trains only on the best one, entropy might decrease but correctness increases. Doesn't this break the collapse theorem?*

Response 2. *This objection confirms rather than refutes our thesis. Consider what “selecting the best solution” requires:*

1. A verifier or reward function that evaluates solution quality.
2. This verifier must encode knowledge of what constitutes a “good” solution.
3. This knowledge was designed by humans.

In AlphaZero [8], the verifier is the game rules—a compact, perfect, human-designed evaluation function. In RLHF [12], the reward model is trained on human preferences. In best-of- N sampling with a code interpreter, the interpreter was human-engineered.

Every filtering mechanism that breaks distributional collapse does so by introducing an external signal. By Definition 1.6, this constitutes human influence. The target theorem explicitly addresses the case where no such external signal exists. If filtering is needed to avoid collapse, then the role and origin of that filtering signal must be accounted for.

Formally, let $\mathcal{R} : \mathcal{V}^* \rightarrow \mathbb{R}$ be a reward function and define the filtered distribution:

$$p_{\theta_k}^{\mathcal{R}}(\mathbf{x}) \propto p_{\theta_k}(\mathbf{x}) \cdot \exp(\beta \cdot \mathcal{R}(\mathbf{x})) \quad (18)$$

The information in $p_{\theta_k}^{\mathcal{R}}$ is bounded by:

$$I(p_{\theta_k}^{\mathcal{R}}; p_{\text{true}}) \leq I(p_{\theta_k}; p_{\text{true}}) + I(\mathcal{R}; p_{\text{true}}) \quad (19)$$

The term $I(\mathcal{R}; p_{\text{true}})$ is strictly positive—the reward function contributes new information about ground truth. But this information originates from human design. Remove the human-designed $\mathcal{R}$ and you are back to unfiltered self-distillation, which collapses by Theorem 3.1.

This collapse has been extensively validated empirically for self-consuming or heavily synthetic training regimes. Alemohammad et al. [16] demonstrate across multiple architectures that self-consuming generative loops cause monotonic increases in FID and decreases in both precision

and recall—only fresh external data prevents this entropy contraction. Dohmatob et al. [17] show that iterative self-training on Llama2-7B causes linear degradation in test loss per generation, with low-probability distributional tails vanishing first—precisely the tail-pruning mechanism of Step 1 above. In a stronger result, Dohmatob et al. [18] prove that even 1% synthetic data contamination triggers collapse, and that larger models amplify rather than mitigate the effect, ruling out the hypothesis that scale alone could overcome distributional collapse. These findings complement the concurrent results of Shumailov et al. [5].

4 Gödelian Limits on Self-Verification

Theorem 4.1 (Impossibility of Complete Autonomous Self-Verification). *Let $\mathcal{M}_k$ denote the Turing machine equivalent of θ_k . The model cannot completely verify that its self-modifications constitute genuine global improvements across all possible inputs.*

Proof. Define $\text{IMPROVE}(k) \iff \mathcal{Q}(\theta_{k+1}) > \mathcal{Q}(\theta_k)$ for any quality metric $\mathcal{Q}$ correlating with progress toward AGI.

Gödel’s First Incompleteness Theorem: $\mathcal{M}_k$, being a formal system capable of arithmetic, contains true statements it cannot prove:

$$\exists \phi_k : \mathcal{M}_k \not\vdash \phi_k \quad \text{and} \quad \mathcal{M}_k \not\vdash \neg \phi_k \quad (20)$$

Statements about whether θ_{k+1} will fail on inputs outside $\text{supp}(\mathcal{D}_k^{\text{self}})$ may be among the undecidable statements.

Gödel’s Second Incompleteness Theorem: The model cannot prove its own consistency:

$$\mathcal{M}_k \not\vdash \text{Con}(\mathcal{M}_k) \quad (21)$$

Rice’s Theorem: Any non-trivial semantic property of the function computed by $\mathcal{M}_{k+1}$ is undecidable by any Turing machine.

We emphasize that these results do not prevent *all* verification—a model can check specific test cases, run benchmarks, and verify particular outputs. What they prevent is *complete* verification: the model cannot guarantee that θ_{k+1} is globally better than θ_k across all possible inputs. Since AGI requires correctness across all computable functions (Definition 1.8), partial verification is insufficient. The model can confirm improvement on tested cases while unknowingly degrading on untested ones—and it cannot determine, from within itself, whether this is happening. $\square$

Corollary 4.2 (The Verification Regress). *If $\mathcal{M}_k$ builds a verifier $\mathcal{V}_k$, then $\mathcal{V}_k$ is itself a formal system subject to the same incompleteness. Verifying $\mathcal{V}_k$ requires $\mathcal{V}'_k$, generating an infinite regress:*

$$\mathcal{M}_k \rightarrow \mathcal{V}_k \rightarrow \mathcal{V}'_k \rightarrow \mathcal{V}''_k \rightarrow \dots \quad (22)$$

Each link in the chain inherits the incompleteness of its predecessor and adds its own statistical error.

Empirical evidence supports a weaker practical concern: current LLMs are unreliable self-verifiers without external feedback. Huang et al. [19] demonstrate that intrinsic self-correction (without external feedback) consistently degrades LLM reasoning performance—the model is more likely to flip a correct answer to incorrect than vice versa. Zheng et al. [20] document systematic self-enhancement bias in LLM-as-judge evaluations, with GPT-4 favoring its own outputs by 10%. Most directly, Stechly et al. [21] show that GPT-4’s self-critique causes performance to *decrease* on reasoning tasks, and that the content of critiques is irrelevant—indicating the model cannot reliably generate useful verification signals about its own work.

4.1 Human Symmetry Objection and Empirical Background

Objection 3 (The Human Symmetry Argument). *If Gödelian limits prevent machines from self-improving, they should equally prevent humans, since humans are also finite systems. But humans clearly self-improve (from pre-linguistic hominids to mathematicians). Therefore the Gödelian argument proves too much.*

Response 3. *This objection identifies a real boundary of the formal argument. The formal proof path does not require proving that humans escape Gödelian limits. It requires only the narrower claim that LLM self-modifications cannot be completely certified by the same formal/computational process that generates them. The following human-brain discussion is therefore empirical and philosophical background, not a theorem in the main proof.*

The LLM is modeled as a formal system. *A finite-precision LLM consists of:*

- *Finite-precision floating-point arithmetic (float16, bfloat16, or float32),*
- *Deterministic matrix multiplications, additions, and nonlinear activations,*
- *Pseudo-random number generation for sampling (algorithmically deterministic given a seed).*

Every operation is computable under the finite-precision implementation model. The entire system is realizable as a Turing machine for purposes of the verification-limit argument.

The human-brain asymmetry is not established here as a formal theorem. *The brain is a quantum-biological system:*

- **Synaptic transmission** *involves probabilistic vesicle release governed by quantum tunneling of Ca^{2+} ions through voltage-gated channels.*
- **Ion channel dynamics** *are inherently quantum mechanical—channel state transitions are stochastic processes with irreducible quantum uncertainty.*
- **Molecular processes** *underlying neural computation (protein folding, neurotransmitter binding, enzymatic activity) operate in the quantum regime.*

If any of these quantum processes contribute to cognitive state transitions, then the brain may have access to genuine ontological randomness—not pseudo-randomness generated by an algorithm, but irreducible indeterminacy as characterized by quantum mechanics. Randomness alone, however, does not solve undecidability or establish reliable access to mathematical truth.

Conditional physical consequence: *Let $\hat{\rho}_t$ be the quantum density operator describing the brain’s state. If the decision process depends on quantum measurement outcomes in a way that cannot be simulated by any effective procedure, then the brain would not be modeled by the same class of deterministic finite-precision computations as deployed LLM inference. This is a physical hypothesis, not a premise discharged by the formal proof.*

The asymmetry:

$$\underbrace{\text{LLM}}_{\text{formal system}} \xrightarrow{\text{Gödel applies}} \text{incomplete, cannot fully self-verify} \quad (23)$$

$$\underbrace{\text{Human brain}}_{\text{quantum-biological}} \xrightarrow{\text{separate empirical question}} \text{not used as a formal premise} \quad (24)$$

This does not require accepting Penrose’s specific microtubule hypothesis [10], though the updated Orchestrated Objective Reduction (Orch OR) review [27] provides a detailed theoretical framework. It requires only the weaker claim that quantum effects at the molecular level

influence neural computation—and recent experimental evidence now supports this across three independent lines of inquiry.

Microtubule quantum effects: Babcock et al. [22] report the first observation of collective quantum optical superradiance in microtubule tryptophan networks at room temperature, involving $> 10^5$ coupled chromophores—an inherently quantum phenomenon. Kalra et al. [23] measure electronic energy migration in microtubules at five times the distance predicted by classical Förster theory, with clinically used anesthetics significantly dampening this quantum transport. Wiest et al. [24] demonstrate bidirectional causality: microtubule-stabilizing drugs delay anesthetic-induced unconsciousness in rats, establishing that the quantum state of microtubules is functionally linked to consciousness.

Nuclear spin effects: Li et al. [25] demonstrate that xenon isotopes with nuclear spin are approximately 30% less potent anesthetics than spin-0 isotopes in mice. Since xenon is chemically inert and the only difference between isotopes is nuclear spin, this constitutes direct evidence that a quantum property modulates consciousness.

Posner molecules: Straub et al. [26] provide the first experimental support for Fisher’s Posner molecule hypothesis, demonstrating lithium isotope-dependent effects on calcium phosphate formation that are “entirely unexpected from classical chemistry” but match quantum dynamical selection predictions (published in PNAS).

These results suggest that quantum effects may be relevant to neural function, but they do not by themselves prove non-computable cognition or an escape from formal limits.

Therefore, the fact that humans can self-improve does not by itself refute the formal self-verification limitation for LLMs. The stronger claim that humans avoid all analogous limits is outside the formal proof path and should be evaluated separately.

5 The SGD Fixed-Point Trap

Theorem 5.1 (Self-Training Converges to a Trivial Fixed Point). *Under self-training, SGD converges to a fixed point where the model reproduces its own output distribution.*

Proof. The self-training loss is:

$$\mathcal{L}_k(\boldsymbol{\theta}) = -\mathbb{E}_{\mathbf{x} \sim p_{\boldsymbol{\theta}_k}} \left[\frac{1}{T} \sum_{t=1}^T \log p_{\boldsymbol{\theta}}(x_t \mid x_{<t}) \right] = H(p_{\boldsymbol{\theta}_k}) + \text{KL}(p_{\boldsymbol{\theta}_k} \parallel p_{\boldsymbol{\theta}}) \quad (25)$$

This is minimized when $p_{\boldsymbol{\theta}} = p_{\boldsymbol{\theta}_k}$ (where $\text{KL} = 0$). Therefore $\boldsymbol{\theta}_k$ is already a global minimum of $\mathcal{L}_k$. SGD converges back to (approximately) $\boldsymbol{\theta}_k$.

The Fisher information matrix captures the effective dimensionality of optimization:

$$\mathbf{F}_k = \mathbb{E}_{\mathbf{x} \sim p_{\boldsymbol{\theta}_k}} \left[\nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x})^{\top} \right] \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \quad (26)$$

As $p_{\boldsymbol{\theta}_k}$ concentrates (Theorem 3.1), $\text{rank}(\mathbf{F}_k)$ decreases. The model’s gradients become confined to a shrinking subspace. Optimization can only rearrange existing capabilities within this subspace, not generate novel ones. $\square$

Remark 5.2. *This formalizes the intuition: a model training on its own output is trying to become better at being itself. The loss function rewards reproducing what it already does. This is not self-improvement; it is self-reinforcement. Furlanello et al. [28] demonstrate this empirically: Born Again Networks—iterative self-distillation with identical architectures—converge to a performance ceiling after initial generations. Mobahi et al. [29] prove that self-distillation progressively reduces the number of basis functions in the learned model, converging to a fixed point of minimal representational complexity—providing the formal mechanism underlying this convergence.*

6 The Pattern Completion Barrier

Theorem 6.1 (Separation Between Pattern Completion and General Reasoning). *There exist computable functions $f \in \mathcal{F}_{\text{AGI}}$ that are not approximable by statistical pattern completion on any finite training set.*

Proof. We exhibit three classes of functions that are computable but unreachable by pattern matching:

(a) **Gödel-undecidable theorems:** For any consistent formal system $\mathcal{F}$ (including the implicit system represented by all mathematical text in $\mathcal{D}_{\text{train}}$), there exist true mathematical statements not provable within $\mathcal{F}$. Their proofs cannot be found by pattern-matching on existing proofs.

(b) **Kolmogorov-complex algorithms [13]:** Let f be a function with Kolmogorov complexity $K(f) > K_{\text{max}}(\mathcal{D}_{\text{train}})$. Then f cannot be reconstructed from patterns in $\mathcal{D}_{\text{train}}$.

(c) **Empirical facts:** Knowledge about the physical world not recorded in $\mathcal{D}_{\text{train}}$ and not logically derivable from it requires observation, which a closed-loop text system cannot perform. $\square$

Objection 4 (Compositionality). *Even if a program has complexity greater than any single training example, it may be derivable by combining patterns. Humans learn math from examples but prove new theorems.*

Response 4. *Compositionality allows recombination of existing patterns, which is a form of search over the space of combinations. This is addressed by the information ceiling (Theorem 2.1): the total information available for composition is bounded by the training data. Composition can extract latent information more efficiently, but it cannot exceed the information-theoretic ceiling. Dziri et al. [30] demonstrate this directly: transformers solve compositional tasks via linearized subgraph matching rather than systematic reasoning, with accuracy collapsing to near-zero at increasing compositional depth and 82.3% of correct final answers containing intermediate errors. Berglund et al. [31] discover the Reversal Curse—LLMs trained on “A is B” achieve near-zero accuracy on “B is A”—a trivial logical deduction that lies outside what autoregressive pattern completion can extract.*

The human analogy remains outside the formal proof path. A human mathematician proving a new theorem may use cognitive processes and external practices that are not captured by finite-context pattern completion, but this manuscript does not require proving that those processes are non-computable.

A transformer is theoretically Turing-complete only with unbounded computation time and unbounded memory. In practice, it has a fixed context window C (typically $\leq 2^{17}$ tokens) and fixed precision, making it a finite automaton, strictly weaker than a Turing machine.

7 The Computational Complexity Barrier ($P \neq NP$)

Theorem 7.1 (Self-Improvement Requires Solving NP-Hard Problems). *Each step of autonomous self-improvement requires the model to solve an optimization problem over its parameter space that is NP-hard in general. If $P \neq NP$, no polynomial-time algorithm (including the model itself) can reliably find optimal or near-optimal solutions.*

Proof. At each self-improvement step k , the model must find:

$$\theta_{k+1}^* = \arg \min_{\theta \in \mathbb{R}^p} \mathcal{L}_{\text{true}}(\theta) \quad (27)$$

where $\mathcal{L}_{\text{true}}$ is the loss relative to ground truth. This is a non-convex optimization problem over $\mathbb{R}^p$ with $p \sim 10^9$ to 10^{12} dimensions. Non-convex optimization over continuous spaces with

discrete structural constraints is NP-hard in general, as is the closely related problem of training neural networks to global optimality.

The key structural feature is the asymmetry between verification and search:

- **Verification** (checking if a given θ_{k+1} improves on θ_k): compute $\mathcal{L}(\theta_{k+1})$ and compare. This requires $O(p \cdot C \cdot L)$ operations—polynomial in model size.
- **Search** (finding a θ_{k+1} that improves on θ_k): explore a space of size $O(2^{p \cdot b})$ (all possible parameter configurations at b -bit precision). In the worst case, this is exponential.

This is precisely the P vs. NP structure: verification is polynomial, but search is (under $P \neq NP$) superpolynomial.

Stochastic gradient descent provides a heuristic that avoids exhaustive search by following local gradients. However, SGD:

1. Converges to local minima, not global minima,
2. Cannot escape basins of attraction without external perturbation,
3. On self-generated data, converges to the trivial fixed point θ_k (Theorem 5.1).

Finding the global optimum—or even a substantially better local optimum—requires solving the NP-hard search problem that SGD explicitly avoids. $\square$

Remark 7.2 (Empirical Observations on LLMs and NP-Hard Problems). *An LLM trained on the entirety of available human text has been exposed to billions of instances of solved combinatorial problems—traveling salesman solutions, scheduling solutions, graph coloring solutions, satisfiability instances, cryptographic challenges. If polynomial-time statistical shortcuts existed for NP-hard problems—that is, if there were learnable patterns in the structure of NP solutions—then a sufficiently large pattern-matching system trained on sufficiently many examples would discover them.*

Current LLMs are very large pattern-matching systems trained on very large corpora. Despite this:

1. *LLMs fail to solve large instances of the traveling salesman problem.*
2. *LLMs cannot factor large semiprimes.*
3. *LLMs cannot find optimal schedules for non-trivial constraint satisfaction problems.*
4. *LLMs cannot reliably solve novel SAT instances beyond small sizes.*

These failures are evidence about current architectures, training methods, and inference procedures. They are not a proof of $P \neq NP$ and should not be used as a theorem-level premise. The formal complexity barrier remains conditional on the relevant optimization problem and complexity assumptions.

Corollary 7.3 ($P \neq NP$ Strengthens the Information Ceiling). *Even information that is technically “latent” in the training data (the information ceiling of Theorem 2.1) may be computationally inaccessible. Under $P \neq NP$, extracting certain latent information requires superpolynomial computation:*

$$\exists f \in NP : \quad f \text{ is derivable from } \mathcal{D}_{\text{train}} \text{ but requires } 2^{\Omega(n)} \text{ steps to extract} \quad (28)$$

Therefore, the effective information available to the model is strictly less than the theoretical information ceiling:

$$I_{\text{accessible}}(\theta_k; p_{\text{true}}) < I(\theta_k; p_{\text{true}}) \leq I(\mathcal{D}_{\text{train}}; p_{\text{true}}) \quad (29)$$

The information ceiling was already a hard limit. Computational complexity makes the effective ceiling even lower.

Remark 7.4 (The Self-Improvement Loop as Iterated Hard Search). *Under the stated complexity assumptions, each iteration of autonomous self-improvement requires solving a hard optimization problem (finding better parameters). The model’s tool for this search is statistical pattern matching. Current failures on NP-hard benchmarks are empirical corroboration of limited search ability, not proof of $P \neq NP$. Fan et al. [32] benchmark 12 LLMs across complexity classes, finding weighted accuracy collapses from 0.25 (NP-complete) to 0.02 (NP-hard). If $P \neq NP$ and the parameter-search problem instantiates an NP-hard family, additional parameters, data, or compute do not by themselves convert polynomial verification into guaranteed superpolynomial search.*

8 The Context-Computation Paradox

Theorem 8.1 (Self-Correction Requires Unbounded Resources). *Let $\theta_k \in \mathbb{R}^p$ be the model parameters at step k , with p on the order of 10^9 to 10^{12} . Let C be the context window size (maximum number of tokens processable in a single forward pass). For the model to verify or correct its own parameters, it must represent and reason about the state space $\mathbb{R}^p$. The following bound holds:*

$$\frac{p \cdot b}{C \cdot \log_2 |\mathcal{V}|} \gg 1 \quad (30)$$

where b is the number of bits per parameter (typically 16 or 32). That is, the information required to describe the model’s own parameter space vastly exceeds the model’s processing capacity in any single reasoning step.

Proof. A model with $p = 10^{11}$ parameters at 16-bit precision contains 1.6×10^{12} bits of state information. With a context window of $C = 128,000$ tokens and vocabulary size $|\mathcal{V}| \approx 10^5$, each context window carries approximately $128,000 \times 17 \approx 2.2 \times 10^6$ bits of information.

The ratio:

$$\frac{\text{model state}}{\text{context capacity}} \approx \frac{1.6 \times 10^{12}}{2.2 \times 10^6} \approx 7.3 \times 10^5 \quad (31)$$

The model would need approximately 730,000 sequential reasoning steps to even *read* its own parameters, let alone reason about them.

For self-correction, the model must:

1. Identify which parameters contribute to errors (requires gradient computation across the full parameter space),
2. Determine correct parameter values (requires optimization over $\mathbb{R}^p$),
3. Verify that corrections do not introduce new errors (requires evaluation across all possible inputs).

Each step requires processing information proportional to p or larger. Since the model can process at most C tokens per step, the number of required steps grows as $\Omega(p/C)$ at minimum. For verification across all inputs (step 3), the required computation is unbounded.

Therefore, even if the model could in principle self-correct (which Theorem 4.1 shows it cannot do completely), the computational resources required grow without bound relative to its finite processing capacity:

$$\lim_{p \rightarrow \infty} \frac{\text{Resources required for self-correction}}{\text{Resources available per step}} = \infty \quad (32)$$

□

Theorem 8.2 (The Overhead Paradox). *Even in the theoretical best case where the context window C is extended to equal the full parameter representation ($C \cdot \log_2 |\mathcal{V}| = p \cdot b$), self-correction remains impossible due to computational overhead that creates an irreducible deficit.*

Proof. Suppose we grant the model a context window large enough to hold its entire parameter state. The context must then be partitioned among competing demands:

$$C = C_{\text{state}} + C_{\text{reasoning}} + C_{\text{temp}} + C_{\text{output}} \quad (33)$$

where:

- C_{state} : tokens representing the parameters to be examined,
- $C_{\text{reasoning}}$: tokens for the logical operations analyzing those parameters (comparisons, gradient estimation, error diagnosis),
- C_{temp} : tokens for intermediate results and working storage,
- C_{output} : tokens for the corrected parameter specifications.

Since $C_{\text{state}} = C$ (by assumption, the model’s full state fills the context), we require:

$$C_{\text{reasoning}} + C_{\text{temp}} + C_{\text{output}} > 0 \quad (34)$$

for any non-trivial reasoning to occur. But $C_{\text{state}} + C_{\text{reasoning}} + C_{\text{temp}} + C_{\text{output}} > C$, which is a contradiction.

In practice, meaningful self-analysis requires substantial reasoning overhead. At minimum, $C_{\text{reasoning}} \geq \alpha \cdot C_{\text{state}}$ for some constant $\alpha > 0$ reflecting the complexity of error diagnosis. Therefore:

$$C_{\text{required}} \geq (1 + \alpha) \cdot C_{\text{state}} + C_{\text{temp}} + C_{\text{output}} > C \quad (35)$$

The deficit is irreducible: *any* system reasoning about a representation of itself requires more resources than the representation alone occupies. $\square$

Theorem 8.3 (The Scaling Trap). *Increasing the context window C to accommodate self-reasoning does not resolve the deficit, because model complexity grows with context capacity.*

Proof. Let C_k be the context window at self-improvement step k , and p_k the parameter count. In practice, model capability scales with both p and C , and the relationship is approximately:

$$p_k = g(C_k), \quad \text{where } g \text{ is monotonically increasing} \quad (36)$$

(larger context windows require more parameters to effectively utilize the additional context—embedding layers scale as $O(C \cdot d)$, attention heads must increase, etc.). Therefore the ratio:

$$R_k = \frac{p_k \cdot b}{C_k \cdot \log_2 |\mathcal{V}|} \quad (37)$$

does not converge to 1 as C_k increases. Empirically, R_k remains large or grows, because $g(C_k)$ grows at least linearly in C_k .

Even setting aside this architectural coupling, the *computational cost* of processing context C in the standard transformer is:

$$\text{FLOPS}_{\text{attention}} = O(C^2 \cdot d) \quad (38)$$

due to the quadratic scaling of self-attention. Doubling C quadruples the compute required per forward pass. The time to perform one self-reasoning step therefore grows as:

$$T_{\text{step}}(C) = O(C^2) \quad (39)$$

For a model attempting to reason about its full parameter space ($C \propto p$), the time per self-correction step is:

$$T_{\text{step}} = O(p^2) \quad (40)$$

For $p = 10^{11}$, this yields $T_{\text{step}} \propto 10^{22}$ FLOPS per reasoning step. At current hardware speeds ($\sim 10^{18}$ FLOPS/s for the largest clusters), a single self-reasoning pass would take approximately 10^4 seconds ≈ 3 hours—for one step of what would require millions of sequential steps to cover the parameter space. $\square$

Theorem 8.4 (Training Time Divergence). *The time required for the model to retrain itself after identifying corrections grows superlinearly with model size, creating a secondary resource explosion.*

Proof. Suppose the model, at self-improvement step k , has identified a set of corrections to apply to its parameters. Implementing these corrections requires a training process (gradient-based optimization). A single training step requires:

$$T_{\text{train_step}} = O(p \cdot C \cdot L) \quad (41)$$

FLOPS, where L is the number of transformer layers. Convergence to a new parameter configuration typically requires S training steps, where S depends on the magnitude of the correction and the loss landscape curvature. The total training time is:

$$T_{\text{train_total}} = S \cdot O(p \cdot C \cdot L) \quad (42)$$

For the self-improvement loop, this training must occur at *every* iteration k . The total compute for K iterations of self-improvement is:

$$T_{\text{total}} = \sum_{k=0}^{K-1} \left[\underbrace{T_{\text{diagnose}}(k)}_{\text{identify errors}} + \underbrace{T_{\text{train}}(k)}_{\text{apply corrections}} + \underbrace{T_{\text{verify}}(k)}_{\text{check results}} \right] \quad (43)$$

By Theorem 5.1, the achievable improvement $\Delta_k \rightarrow 0$, meaning each successive iteration produces diminishing returns. But the *cost* per iteration does not diminish—it remains $\Omega(p \cdot C \cdot L)$ per training cycle. The ratio of cost to benefit therefore diverges:

$$\frac{T_{\text{train}}(k)}{\Delta_k} \rightarrow \infty \quad \text{as } k \rightarrow \infty \quad (44)$$

Furthermore, as errors accumulate (Theorem 9.1), the corrections required become more complex (affecting larger portions of the parameter space), increasing S_k . Simultaneously, the model’s ability to correctly diagnose errors degrades (since it is using a progressively more corrupted model to identify corruption). This creates a feedback loop:

1. More errors $\implies$ more parameters need correction $\implies$ longer training,
2. Longer training on self-generated data $\implies$ more distributional collapse (Theorem 3.1),
3. More collapse $\implies$ worse error diagnosis $\implies$ more errors introduced.

The self-improvement process is therefore not merely slow but *anti-convergent*: the resources required per unit of improvement grow without bound, while the quality of each improvement attempt simultaneously degrades. In the limit:

$$\lim_{k \rightarrow \infty} \frac{\text{Compute cost of iteration } k}{\text{Genuine improvement from iteration } k} = \infty \quad (45)$$

Under these assumptions, even with unbounded time and compute, the process does not converge—it diverges, because the errors introduced at each step eventually exceed the corrections achieved. $\square$

Remark 8.5 (The Infinite Loop). *Theorems 8.1–8.4 together describe an inescapable recursive trap:*

1. *The model cannot fit a representation of itself into its working memory (8.1).*
2. *Even if it could, the overhead of reasoning about that representation exceeds the available space (8.2).*
3. *Expanding the space increases the model’s complexity proportionally, preserving the deficit (8.3).*
4. *The time to retrain after each correction cycle diverges relative to the improvement gained (8.4).*

Under the resource-scaling assumptions above, every attempt to provide the system with sufficient resources for self-correction generates a new, larger resource requirement. The gap between what the system needs and what it has is not a fixed deficit to be closed by engineering—it is a structural property that scales with any attempt to close it. Combined with the conditional error model of Theorem 9.1, this creates an anti-convergent loop rather than a guaranteed path to self-correction.

Empirically, Liu et al. [33] demonstrate that LLMs exhibit a U-shaped performance curve over context position, with significant degradation for information in the middle—if a model cannot reliably access information throughout its own context, the comprehensive self-analysis required for self-improvement is infeasible. The scaling trap is quantified by Kaplan et al. [34], who show language model loss follows power-law scaling with exponents of ~ 0.07 for parameters, guaranteeing diminishing returns where each $10\times$ compute increase yields only $\sim 12\%$ loss reduction. Hoffmann et al. [35] confirm this with compute-optimal training: Chinchilla (70B) outperforms Gopher (280B) by only 7% on MMLU despite vastly more efficient resource allocation.

9 Error Compounding in Recursive Self-Modification

Theorem 9.1 (Conditional Error Divergence Without External Correction). *Let ϵ_k measure the error of θ_k relative to ground truth. Assume the recursive self-training process satisfies:*

$$\epsilon_{k+1} = \epsilon_k - \Delta_k + \delta_k + \sum_{j=0}^k \alpha_{k,j} \delta_j \quad (46)$$

where $\Delta_k \geq 0$ is the intended improvement, $\delta_k > 0$ is undetected degradation, and $\alpha_{k,j} \geq 0$ captures error propagation through self-generated training data.

If errors propagate through the data chain, $\Delta_k \rightarrow 0$, and δ_k remains bounded away from zero, then:

$$\liminf_{k \rightarrow \infty} \epsilon_k = \infty \quad (47)$$

The error diverges under these assumptions. Without these assumptions, the theorem should be read as a recurrence analysis rather than a universal claim about every possible autonomous process.

Remark 9.2 (The Statistical Error Floor). *The fundamental issue is that the system is stochastic. Every output has an associated error probability. When the system uses its own outputs to train itself, it trains on data that contains errors. The next generation inherits those errors and adds its own. The error is cumulative and the correction mechanism (the model itself) has the same statistical error rate as the system it is trying to correct. This is not a bug to be fixed; it is a mathematical property of using statistics to correct statistics.*

An external corrector (human or human-designed verifier) breaks this cycle because its error properties are independent of the model’s. But an AI-generated corrector has correlated errors—it

was built by the same flawed process—and therefore cannot provide the independent error correction needed. Gao et al. [36] establish scaling laws for reward model overoptimization: true quality initially improves then degrades as optimization against any proxy intensifies, demonstrating that even human-designed reward models degrade under iterative optimization—the problem would be far worse with self-generated evaluation. A concurrent formal analysis [37] independently models recursive self-training as a discrete-time dynamical system, proving entropy decay and variance amplification invariants that hold regardless of model architecture.

10 Main Theorem

Theorem 10.1 (Conditional Impossibility of Autonomous LLM Self-Improvement to AGI). *Let $\{p_{\theta_k}\}_{k=0}^{\infty}$ be the sequence of models produced by autonomous self-improvement (Definition 1.7). Assume the AGI nontriviality condition (Assumption 1) and the eight barriers listed below hold under the formal definitions of the model, training process, and AGI criterion. Then $\{p_{\theta_k}\}$ cannot converge to AGI (Definition 1.8) unless the initial model already satisfies the AGI criterion.*

1. **Information ceiling** (Theorem 2.1): *Under the Markov-chain assumptions, self-generated data cannot create new mutual information about ground truth. Inference-time search can better utilize existing information, so the proof must distinguish information creation from computational extraction.*
2. **Distributional collapse** (Theorem 3.1): *Under finite-sample, unsmoothed, unfiltered empirical self-distillation, self-training contracts empirical support and can reduce entropy. Mechanisms that prevent collapse must be analyzed for whether they introduce external information.*
3. **Gödelian verification failure** (Theorem 4.1): *The model cannot completely verify its self-modifications as genuine global improvements. Partial verification (benchmarks, test cases) is possible. The human-brain asymmetry discussion is empirical background rather than a formal premise.*
4. **SGD fixed-point trap** (Theorem 5.1): *Self-training converges to reproducing the model’s own distribution. The Fisher information rank contracts, confining optimization to a shrinking subspace.*
5. **Pattern completion barrier** (Theorem 6.1): *The target claim is that stochastic pattern completion on finite training data is weaker than general computation under precise finite-model assumptions. Compositionality enables recombination but remains bounded by the information ceiling. Practical transformers are finite automata, not Turing machines.*
6. **Computational complexity barrier** (Theorem 7.1, Corollary 7.3, and Remark 7.2): *If the relevant self-improvement search problem instantiates an NP-hard optimization family and $P \neq NP$, then the gap between verification and search is permanent. Current LLM failures on NP-hard benchmarks are empirical context, not proof of $P \neq NP$.*
7. **Context-computation paradox** (Theorems 8.1–8.4): *The model’s working memory is orders of magnitude too small to represent its own parameter space. Even with unlimited context, computational overhead for reasoning about the representation exceeds available space (Theorem 8.2). Expanding resources increases model complexity proportionally, preserving the deficit (Theorem 8.3). Training time per correction cycle diverges relative to improvement gained, and the process becomes anti-convergent: more errors require longer correction, longer correction causes more collapse, more collapse causes worse error diagnosis (Theorem 8.4). The process forms an infinite loop with no halting condition.*

8. **Error divergence** (Theorem 9.1): Under the stated recurrence, lower-bound, propagation, and vanishing-improvement assumptions, statistical errors compound through recursive self-training. The universality of those assumptions remains a proof obligation.

Under these assumptions, the target conclusion is:

$$\boxed{\forall \theta_0 \in \mathbb{R}^p, \forall k \geq 0 : \mathcal{S}^k(\theta_0) \not\rightarrow \theta_{\text{AGI}}^*} \quad (48)$$

within the autonomous stochastic autoregressive framework, excluding the degenerate case in which θ_0 already has AGI-level information. The remaining work is to discharge the assumptions above in a machine-checkable form.

11 Discussion

11.1 What the Proof Program Targets

The proof program targets the following conditional result: if the information ceiling, collapse, verification, fixed-point, complexity, resource, and error-divergence barriers are formalized under the definitions above, then a stochastic language model operating without any form of human influence cannot self-improve to AGI unless it already satisfies the AGI criterion. The intended scope is broad with respect to model size, context length, and compute budget, but those scaling claims remain part of the verification work.

11.2 What Would Escape These Bounds

A system escapes these bounds if it incorporates:

- External information sources (new data, environment interaction) — violates closed-loop assumption,
- Human-designed evaluation signals (reward functions, preference models) — constitutes human influence,
- Human-engineered tools or verifiers (code interpreters, proof checkers, game rules) — constitutes human influence,
- Non-stochastic components (formal theorem provers, symbolic reasoning engines) — departs from the stochastic framework.

Each of these either introduces human influence or leaves the stochastic paradigm. In either case, the system is no longer what the singularity hypothesis describes: an AI that autonomously bootstraps itself to superintelligence without human involvement.

11.3 The Stochastic Parrot, Formalized

The proof program aims to provide mathematical grounding for the characterization of LLMs as “stochastic parrots” [6]. The model is a conditional probability estimator: $p_{\theta}(x_t | x_{<t})$. Self-improvement within this framework is self-distillation: the model learning to reproduce its own distribution. Under the information-ceiling assumptions, the output may become more fluent, more consistent, or more confident without becoming more correct about things absent from training or more general than the information-theoretic ceiling permits.

11.4 Human Influence as Theorem Target

The central theorem target is not that LLMs cannot improve—they clearly can and do. It is that *human influence is a necessary component of open-ended improvement* under the autonomous stochastic-language-model framework. Whether through curated training data, designed reward functions, engineered tools, or architectural innovations, current LLM progress depends on external structure. The formal task is to show that severing this dependency requires escaping at least one of the assumptions behind the barriers above. Until those assumptions are discharged, this should be read as a conditional impossibility claim rather than a completed theorem.

11.5 A Control Theory Perspective

Control theory provides an intuitive framing for the central impossibility. A feedback control system works because the controller is *independent* of the plant. A thermostat is not made of the same heat it regulates.

An LLM attempting to regulate itself is simultaneously the plant and the controller. The sensor, actuator, and error estimator are all the same stochastic process with the same systematic biases. When the system drifts, the component detecting the drift has drifted by the same amount in the same direction.

In control theory terms, the controller’s noise is coupled to the plant’s noise with correlation approaching 1. This violates the fundamental assumption that makes feedback control work: independent error correction.

More precisely, classical control theory requires four prerequisites that LLMs violate:

1. **Observable state.** The meaningful state of an LLM is the distribution p_{θ} over all possible inputs—a function over an exponentially large space. This state is not observable; it can only be sampled. No finite number of samples characterizes the distribution.
2. **Plant model.** The relationship between parameter changes $\Delta\theta$ and behavioral changes Δp_{θ} is nonlinear, non-convex, and not characterizable in closed form. No transfer function exists. Without a plant model, no controller can compute corrective inputs.
3. **Measurable error signal.** Feedback control requires $e(t) = r(t) - y(t)$, where $r(t)$ is a reference signal. The reference signal for “general intelligence” would require a complete specification of AGI—which loops back to the verification problem of Theorem 4.1.
4. **Stability guarantees.** Lyapunov stability and BIBO stability require structural properties (linearity, bounded gain, known dynamics) that LLM training dynamics lack. The self-training loop has no Lyapunov function guaranteeing convergence.

A PID controller on a motor works because you can measure the angle, you know the transfer function, and the error signal is trivially defined. An LLM fails on all three. The system is unobservable, uncharacterizable, and has no well-defined reference signal. Self-regulation under these conditions is not an engineering problem to be solved—it is a category error.

11.6 The Multi-Agent Correlation Problem

Objection 5 (Multiple Agents Checking Each Other). *What if we deploy multiple LLM agents in a pipeline—an “architect” that diagnoses problems and a “programmer” that implements fixes, or a “generator” and a “critic”—so that they provide independent error correction for each other?*

Response 5. *This does not work because the agents’ errors are correlated. Multiple agents derived from the same (or similar) training distributions share:*

- *The same training data $\implies$ the same blind spots,*

- The same pretraining objective $\implies$ the same systematic biases,
- The same failure modes on edge cases and distribution-boundary inputs.

Let δ_{gen} be the error introduced by the generator and δ_{crit} be the error in the critic’s evaluation. For independent correction, we need:

$$\text{Corr}(\delta_{\text{gen}}, \delta_{\text{crit}}) \approx 0 \quad (49)$$

But since both agents are functions of the same training distribution $\mathcal{D}_{\text{train}}$:

$$\text{Corr}(\delta_{\text{gen}}, \delta_{\text{crit}}) \rightarrow 1 \quad \text{as overlap in training data} \rightarrow 1 \quad (50)$$

When the generator introduces a subtle bug, the critic has the same blind spot because it learned from the same distribution. The errors resonate—they reinforce rather than cancel.

This generalizes to any number of agents: n agents with correlated errors provide strictly less than n independent checks. In the limit of identical training distributions, n agents provide exactly one independent perspective, regardless of n . The correlation is not in the architecture—it is in the training distribution. You cannot escape it by factoring the agent into components.

The SWE-CI benchmark [7] provides direct empirical evidence: its dual-agent architecture (Architect + Programmer) still exhibits regressions in over 75% of tasks despite the agents checking each other (see Section 11.8). Kim et al. [38] evaluate over 350 LLMs and find 60% error agreement when both models err, with error correlation increasing for larger, more accurate models—demonstrating that scaling amplifies rather than alleviates the shared blind spots that prevent reliable mutual verification.

The only way to decorrelate errors is to introduce a signal from outside the distribution—which is human influence by Definition 1.6.

11.7 Why Noise Cannot Substitute for Human Influence

A natural objection to the multi-agent correlation problem is: “Add randomness to decorrelate the agents.” This fails because of Theorem 2.1.

White noise is statistically independent of the model’s errors but carries zero mutual information with ground truth:

$$I(\text{noise}; p_{\text{true}}) = 0 \quad (51)$$

Adding noise to the model’s output:

$$I(p_{\theta_k} + \text{noise}; p_{\text{true}}) \leq I(p_{\theta_k}; p_{\text{true}}) \quad (52)$$

by the data processing inequality. Information is *lost*, not gained. The noise decorrelates the agents’ errors in random directions, not toward correctness. The result is agents that disagree more often, but not agents that disagree *usefully*.

What is needed is not independence alone—it is *informed* independence. The external corrector must have mutual information with ground truth that is statistically independent of the model’s errors. A human reviewer satisfies this: their knowledge originates from a completely different process (biological cognition, real-world embodied experience) and correlates with ground truth through a different channel.

In control theory terms (Section 11.5): adding noise to the sensor does not improve the state estimate. What is needed is a *second sensor* that measures the actual system through a different physical principle. Human oversight is that second sensor.

11.8 Empirical Corroboration: The SWE-CI Benchmark

Independent empirical evidence for the theoretical results of this paper is provided by SWE-CI (Software Engineering—Continuous Integration) [7], the first repository-level benchmark evaluating LLM agent capabilities in maintaining codebases over iterative CI loops.

11.8.1 Experimental Setup

SWE-CI comprises 100 tasks drawn from 68 real-world repositories, with each task spanning an average of 233 days and 71 consecutive commits of authentic evolutionary history. The benchmark evaluates 18 models from 8 providers, consuming over 10 billion tokens. The iteration cycle—run tests, define requirements, modify code, repeat—instantiates precisely the autonomous self-improvement loop of Definition 1.7, applied to code rather than model weights.

SWE-CI employs an Architect–Programmer dual-agent protocol: the Architect identifies functional gaps and issues requirements; the Programmer implements them. This is an explicit attempt at independent error correction through role decomposition.

11.8.2 Key Result: Zero-Regression Rate

The zero-regression rate measures the fraction of tasks in which the agent never regresses—that is, never breaks a previously passing test throughout the entire maintenance process. This metric directly corresponds to whether $\delta_k > 0$ (undetectable degradation) occurs in our error model (Theorem 9.1).

Result: Most models achieve a zero-regression rate below 0.25. In over 75% of tasks, the agent introduces at least one regression. Only two models in the Claude Opus series exceed a zero-regression rate of 0.5, meaning even the best models regress in roughly half of all tasks. This is a direct, large-scale measurement of $\delta_k > 0$.

11.8.3 Mapping to Theoretical Results

Theorem	SWE-CI Evidence
9.1 (Error Divergence)	Zero-regression rate < 0.25 for most models: regressions occur in 75%+ of tasks, directly measuring $\delta_k > 0$
3.1 (Distributional Collapse)	Agent quality degrades over longer iteration sequences; EvoScore with $\gamma > 1$ penalizes models that accumulate technical debt
Sec. 11.6 (Correlated Corrector)	Dual-agent architecture (Architect + Programmer) still fails at comparable rates—role decomposition does not provide independent error correction
Sec. 11.5 (Human Influence)	Model improvement across the evaluation period comes only from human-driven releases (new architectures, training data, RLHF), not from autonomous iteration

Table 1: Mapping between theoretical impossibility results and SWE-CI empirical observations.

11.8.4 Within-Run Degradation vs. Across-Release Improvement

SWE-CI’s results reveal a striking asymmetry. *Within* an autonomous run (the CI iteration loop), quality degrades: regressions accumulate, technical debt compounds, and the agent’s ability to maintain the codebase deteriorates over successive iterations. *Across* model releases (Claude Sonnet to Claude Opus, GPT-4o to GPT-o3, etc.), performance improves steadily.

This asymmetry is consistent with what the theory predicts. Autonomous iteration (no human influence) leads to error divergence (Theorem 9.1) and distributional collapse (Theorem 3.1). Inter-release improvement reflects human-driven development: new architectures, curated training data, human-designed reward functions, and engineering effort. The intelligence driving progress across releases originates in human cognition, supporting the theorem target in Section 10.4.

The SWE-CI benchmark thus provides empirical corroboration, across 18 models and 100 tasks, for several failure modes targeted by this proof program [7, 5].

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